Machine Learning Algorithms and Applications

The Machine Learning Process

1. Data Pre-Processing

- Import the data
- Clean the data
- Split into training & test sets
- Feature Scaling

2. Modelling

- Build the model
- Train the model
- Make predictions

3. Evaluation

- Calculate performance metrics
- Make a prediction

Feature Scaling

What is it?

- Adjusts input variables to a comparable range
- Uses standardization or normalization

Why Do We Need It?

- Better results: KNN, SVM, and Gradient Descent depend on distance, so scaling improves their performance
- Faster optimization: Models reach a solution more quickly when features share a similar scale
- Fairer weighting: Large values no longer overpower smaller ones

Example

- Height (cm) vs weight (kg) often have very different scales
- Without scaling, one can unfairly influence the model

Feature Scaling: Normalization vs Standardization

Normalization

- Maps values to the interval [0, 1]
- Works well for distance-based models (e.g., KNN)
- **Formula:** $X' = (X - X_{min}) / (X_{max} - X_{min})$

Standardization

- Centers data at 0 with unit variance
- Fits models that assume normal distribution (e.g., Logistic Regression)
- **Formula:** $X' = (X - \mu) / \sigma$

Feature Scaling: Choosing the Right Method

Normalization

- Best when features span different ranges and you want them in [0, 1]
- Example: Min-Max Scaling

Standardization

- Best when you need a bell-shaped distribution or the data has outliers
- Centers data and scales by standard deviation

Feature Scaling: Example

Before Scaling

- Height: 160, 170, 180 cm
- Weight: 55, 70, 90 kg
- Different scales (150–190 vs 45–120)

After Min-Max (0 to 1)

- Height: 0.25, 0.50, 0.75
- Weight: 0.13, 0.33, 0.60
- Same range, fair contribution

Categorical Data

What is Categorical Data?

- Values that represent groups or labels
- **Ordinal:** Ordered levels (e.g., Budget, Mid-range, Premium)

- **Nominal:** No order (e.g., Riyadh, Jeddah, Dammam)

Why Encode It?

- ML models work with numbers only
- Raw: Riyadh, Jeddah, Dammam
- Encoded: [1, 2, 3] or One-Hot

Categorical Data: How to Deal with It?

Ordinal Encoding

Example

- Price tier: Budget, Mid-range, Premium
- Encoded: 0, 1, 2
- Keeps the order

Another Example

- Rating: Poor, Fair, Good, Excellent
- Encoded: 0, 1, 2, 3

Categorical Data: How to Deal with It?

One Hot Encoding

Example

- City: Riyadh, Jeddah, Dammam
- Riyadh → [1, 0, 0]
- Jeddah → [0, 1, 0]
- Dammam → [0, 0, 1]

Why One Hot?

- No implied order between categories
- Each category gets its own column
- Model treats each category equally

Categorical Data: How to Deal with It?

Dummy Encoding

Example

- Region: North, South, East
- North → [1, 0]
- South → [0, 1]
- East → [0, 0] (reference)

Difference from One Hot

- Drops one category as reference
- Fewer columns (avoids multicollinearity)
- Reference category = all zeros

Categorical Data: How to Deal with It?

Other Type Encoding

Label Encoding

- Assigns a unique number to each category
- No new columns created
- May imply order (use with care)

Target Encoding

- Replaces category with mean of target
- Useful for high-cardinality features
- Risk of overfitting

Frequency Encoding

- Replaces category with its count in data
- Captures category importance

Handling Missing Data

What is Missing Data?

- Empty or incomplete values in the dataset
- Can cause biased models or wrong predictions

Types

- **MCAR:** Missing Completely at Random
- **MAR:** Missing at Random
- **NMAR:** Not Missing at Random

Impact

- Skews model results
- Can lead to prediction errors

Handling Missing Data: MCAR

Missing Completely at Random

- Data is missing by chance
- No link to other variables
- No pattern in what is missing

Example

ID	Age	Income	Resp
1	25	5000	Yes
2	30	?	No
3	?	7000	Yes
4	45	9000	?

? appears randomly

MCAR: Limitations & Practical Approach

Why MCAR Is Rare

- Missing data often has causes (human behavior, survey errors, external events)
- Some groups skip answers more than others
- Patterns in missingness are common

Before Assuming MCAR

- Understand how the data was collected
- Consult domain experts
- Work with data engineers

Practical Approach

- If MCAR holds after analysis, use simple imputation
- Otherwise, consider MAR or NMAR methods

Handling Missing Data: MAR

Missing at Random

- Missingness can be explained by other observed features
- The probability of missing depends on what we already have in the data

Example

ID	Age	Income	Resp
1	25	?	Yes
2	28	?	No
3	45	7000	Yes
4	50	9000	No

Income ? when Age is low (pattern)

MAR: Details & Imputation

Key Points

- Missingness depends on observed features, not the missing one
- MAR is more common in real data than MCAR
- Check if missingness changes with other variables

Example

- Survey: students with high grades skip reporting study hours
- Grades (observed) explain why study hours are missing

Imputation for MAR

- kNN Imputation
- Miss Forest
- Use observed features to fill missing values

---

kNN Imputation: Before & After

Before

ID	Age	Income
1	25	5000
2	30	?
3	28	6000

4	32	5500

ID 2: Income missing

After (k=2)

ID	Age	Income
1	25	5000
2	30	5750
3	28	6000
4	32	5500

ID 2: (6000+5500)/2 = 5750

Miss Forest

What is Miss Forest?

- Uses Random Forest to impute missing values
- Iterative: fills missing, refits model, repeats
- Works well with mixed data (numeric + categorical)

How It Works

- 1. Start with simple imputation (e.g., mean)
- 2. Train Random Forest on each feature with missing values
- 3. Predict and fill missing values
- 4. Repeat until convergence

When to Use

- Non-linear relationships
- Multiple missing features
- Often outperforms kNN for complex data

Miss Forest: Example

Before

ID	Age	Income	Score
1	25	5000	70
2	30	?	85

3	28	6000	?
4	32	5500	90

Income & Score missing

After (Miss Forest)

ID	Age	Income	Score
1	25	5000	70
2	30	5650	85
3	28	6000	82
4	32	5500	90

RF predicts from Age, Income, Score

Handling Missing Data: MNAR

Missing Not at Random

- The hardest case of the three
- Missingness is tied to the missing value itself
- Or to features we did not collect

Example

ID	Age	Salary
1	25	8000
2	30	?
3	28	7500
4	35	?

Low salary → people skip (value itself causes ?)

MNAR: Details & Approach

Key Difference

- MCAR: no pattern
- MAR: related to observed features
- MNAR: tied to the missing variable itself

Example

- Health survey: people with high stress avoid reporting it
- Stigma causes non-random missingness

Challenge & Solution

- Hard to fix without more data or domain knowledge
- Add a binary flag: was this value imputed?
- Model can learn from the missingness pattern

Data Preprocessing Template

Steps to Follow

- 1. Load the data
- 2. Check for missing data and handle it
- 3. Encode categorical data
- 4. Split into training and test sets
- 5. Feature scale numerical data

Pipeline

- Raw Data → Handle Missing → Encode Categories
- → Train/Test Split → Scale Features
- → ML Algorithm

Hands-On Code

- Data Preprocessing Template
- Apply the steps we covered